

PRODUCT CASE STUDY

AI-Powered Athlete Content Studio

How a full-stack generative AI platform transforms the way sports brands produce athlete imagery — from weeks of production to seconds.

PRODUCT

Pluribus Studio

CATEGORY

Generative AI · Sports Tech

STAGE

Live — Production

STACK

React · TypeScript · Supabase

AI MODELS

FLUX · Kling · Claude

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Executive Summary

Pluribus is a generative AI content studio built specifically for the sports industry. It gives brands, agencies, and athlete management companies the tools to produce photorealistic campaign imagery — in any scenario, wearing any kit, in any environment — without a physical shoot.

"A platform where you select an athlete, choose a visual recipe, and receive print-ready campaign imagery within seconds — with full identity fidelity and brand consistency."

The platform combines structured photographic recipes, a wardrobe customisation system, an identity-preservation pipeline, and a collaborative campaign workspace into a single end-to-end product. Every design decision was made to serve non-technical users — creative directors, brand managers, and sports marketers — not engineers.

~5s

Draft image generation time

15+

Curated campaign recipe templates

6

AI models integrated

100%

Serverless, zero-infra operation

The Problem

Professional sports content production is expensive, slow, and inflexible. A single athlete campaign shoot — booking the athlete, venue, photographer, lighting crew, post-production, and retouching — routinely costs £30,000–£100,000 and takes four to eight weeks from brief to delivery. For brands managing rosters of 10–100 athletes across multiple campaigns per year, this is operationally unsustainable.

The secondary problem is consistency. When an athlete's availability changes, a sponsor changes, or a brief evolves mid-campaign, the cost of reshooting even a single image is prohibitive. Brands are forced to make do with existing footage or accept compromised outputs.

TRADITIONAL WORKFLOW PAIN POINTS

- 6–8 week turnaround per shoot
- £30k–100k per campaign
- Athlete scheduling conflicts
- No iteration once shot is done
- Inconsistent look across athletes
- Re-shoot cost on brief changes

WHAT BRANDS ACTUALLY NEED

- Same-day turnaround for approvals
- Unlimited creative iterations
- Consistent visual identity
- Kit and wardrobe flexibility
- Any scenario, any environment
- One platform for the whole team

Pluribus replaces the physical shoot with a structured AI workflow. Rather than exposing raw AI prompting to users — which produces inconsistent, unpredictable results — the platform encodes photographic knowledge into **Campaign Recipes**: structured templates that specify camera angle, lighting setup, background, wardrobe, pose, expression, and motion, then translate those parameters into a precise generation prompt automatically.

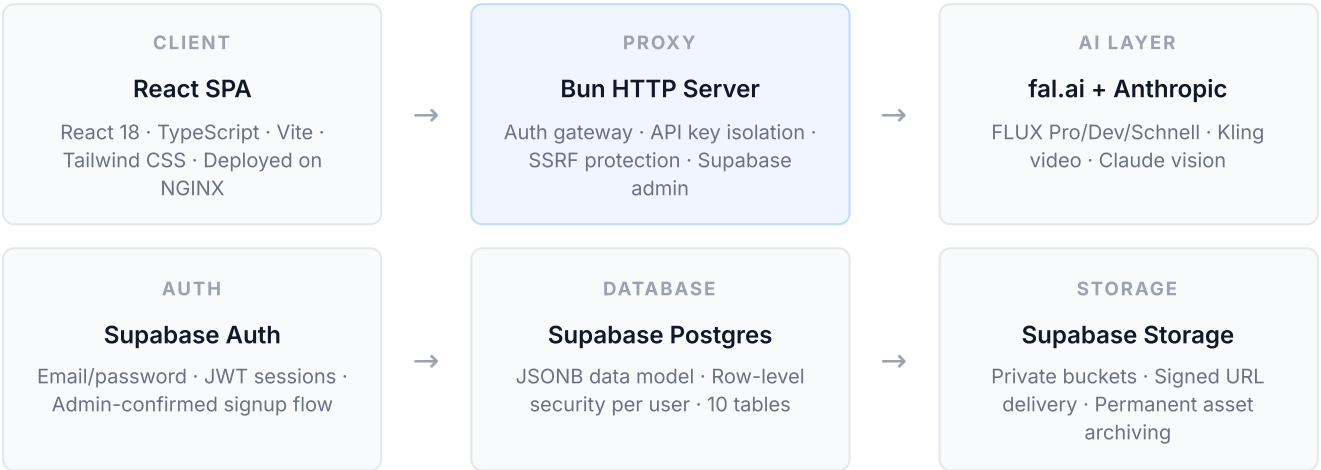
Identity fidelity is handled through a **Capture Profile** system. Each athlete is given a reference library of angles and expressions. The platform uses Claude AI to analyse uploaded photos and automatically infer physical attributes — build, skin tone, hair — creating a consistent identity constraint set that is injected into every generation. The result is outputs that are recognisably the same person across different scenarios.

The core insight: the bottleneck in AI image generation for brands is not the model — it is the prompt. Pluribus removes prompt writing entirely. The creative brief becomes a recipe; the recipe becomes a prompt; the model delivers the image.

Core Product Modules

MODULE	WHAT IT DOES	WHY IT MATTERS
Athlete Profiles	Store reference photos, physical attributes, constraints, and approved/rejected likeness history per athlete	Ensures generational consistency — the same person, every time
Campaign Recipes	15 curated photographic recipe templates across Portrait, Motion, Editorial, Social, Product and Documentary categories with per-field randomisation	Removes the creative knowledge barrier — any team member can produce professional results
Wardrobe System	Visual kit builder with sport-specific clothing types, 18-colour palette, logo placement — or upload a garment flat-lay for Claude vision analysis	Enables full sponsor and kit customisation without a physical wardrobe
Studio	Generation workspace: select athlete + recipe or custom prompt + wardrobe, choose quality tier, generate and review	End-to-end generation in a single screen; no technical knowledge required
Campaign Workspace	Organise outputs by campaign, review generated assets, track resemblance scores, download or share for stakeholder review	Connects generation to the broader production workflow
Stakeholder Review	Shareable review links with no login required; time-limited, read-only access to campaign galleries	Shortens the approval loop from days to hours

Pluribus is built as a single-page application with a thin security proxy. The architecture prioritises zero operational overhead — no Kubernetes, no container orchestration, no custom model hosting. All compute is serverless or third-party managed.



Key Engineering Decisions

DECISION	RATIONALE
In-memory cache with write-through persistence	All store reads are synchronous and instant — no loading spinners for data already on device. Supabase is written to in the background. localStorage provides an offline baseline.
JSONB data model in Postgres	Schema evolution is free — adding fields to Athletes, Recipes, or Kits requires no migrations. The typed TypeScript layer provides safety at the application level.
Bun proxy (not Next.js / Vercel functions)	A single persistent Bun process handles auth validation, API key proxying, and server-side operations with ~0ms cold start. No function timeout limits on long-running generation requests.
Structured recipes over raw prompts	Raw prompts produce inconsistent results and require AI expertise. Structured parameters with human-readable label maps produce deterministic, reproducible creative outputs.
Claude vision for auto-population	Adding an athlete requires only a photo upload. Claude infers build, skin tone, hair, age, and sport — reducing manual data entry to just name and event, accelerating onboarding.
Resemblance scoring	Each generated image is automatically scored against the athlete's reference photo using a perceptual similarity algorithm, giving teams an objective quality signal before approval.

Security Model

API keys (fal.ai, Anthropic) never touch the browser. All AI calls route through the Bun proxy, which validates a Supabase JWT before forwarding. SSRF protection on the storage mirror endpoint restricts CDN URLs to

known fal.ai domains. Row-level security in Postgres ensures no cross-user data access at the database layer.

The guiding principle throughout Pluribus's design is **progressive disclosure**: the simplest path to a great result requires the fewest decisions, while power users have access to full parameter control. A user can generate a professional campaign image in three clicks — select athlete, select recipe, generate — or spend ten minutes configuring lighting direction, crop, motion type, and wardrobe down to the accent colour.

The Recipe System — Photographic Knowledge as Product

The Campaign Recipe is the product's most important design decision. Each recipe encodes the knowledge of a professional photographer into a structured data type spanning 14 parameters across camera, lighting, background, styling, subject direction, motion, and output format. Fifteen curated starter recipes — from "Premium Sports Campaign" to "Cinematic Night Exterior" — are available to all users immediately.

RECIPE PARAMETERS

- Shot type (close-up → full body)
- Lens (35mm → 100mm telephoto)
- Camera angle and crop
- Lighting setup and direction
- Contrast level
- Background type and detail level
- Wardrobe and colour palette
- Pose, expression, energy
- Motion type and intensity
- Output aspect ratio and quality

RANDOMISATION ENGINE

- Per-field randomisation toggle
- Three strength levels (low / medium / high)
- Sport-aware constraints (e.g. pool backgrounds stay pool)
- Identity fields always protected
- Produces creative variation on each run
- "Preview roll" in library before generating

Wardrobe as a Design Surface

The Wardrobe system treats clothing as a creative variable, not a prompt annotation. Users build kits visually — selecting sport-appropriate clothing types from a grid, applying 18-colour swatches to each item, setting fitted/standard/loose fit, and placing a sponsor logo with a placement selector. The platform translates this into a structured English wardrobe description that is appended to the generation prompt: *"Wearing: royal blue fitted running singlet with gold accents, black compression shorts. Nike logo on left chest."*

Alternatively, users upload a flat-lay, product photo, or mockup of the actual garment. Claude vision analyses the image and produces a generation-ready description automatically.

AI Integration — Claude as a Co-worker, Not a Feature

Claude is embedded in three distinct workflows, each invisibly reducing manual effort:

WORKFLOW	WHAT CLAUDE DOES	USER EXPERIENCE
Athlete onboarding	Analyses reference photo via vision API; returns build, skin tone, hair, estimated age, sport	Fields auto-populate 2–3 seconds after upload; user reviews and confirms
Garment analysis	Analyses flat-lay or product photo; returns a concise, generation-ready clothing description	Upload a garment image; get a prompt description instantly; edit if needed
Prompt enhancement	Rewrites and expands a custom prompt typed by the user into a more detailed, generation-optimised brief	Type a rough note; click "Enhance with AI"; receive a polished prompt

Build Journey & Key Learnings



PHASE 1 — FOUNDATION

Auth, data model, and core generation loop

Supabase auth with admin-confirmed signup, athlete profiles, localStorage-backed store with Supabase write-through, first fal.ai FLUX integration with a simple prompt textarea.



PHASE 2 — IDENTITY PIPELINE

Reference capture, constraint injection, and resemblance scoring

Multi-angle capture system with Supabase Storage archiving. Profile prompt constraints auto-injected into every generation. Perceptual resemblance scoring against reference photo.



PHASE 3 — CAMPAIGN WORKSPACE

Project management, runs tracking, and stakeholder review

Campaign galleries, run history with status tracking, shareable review links with time-limited tokens and no login required for stakeholders.



PHASE 4 — RECIPE SYSTEM

Structured photographic recipes replacing raw prompts

15 curated recipes with 14 structured parameters, prompt builder translating parameters to generation text, per-field randomisation engine, full gallery and edit UI.



PHASE 5 — WARDROBE + AI AUTO-FILL

Kit builder, garment upload, Claude vision integration

Sport-specific clothing system with visual colour picker, logo placement, flat-lay analysis via Claude vision. Athlete onboarding auto-population from reference photo.

Hardest Problems Solved

PROBLEM	SOLUTION
Identity consistency across generations	Structured constraint extraction from athlete profiles — specific physical descriptors injected with every prompt, not left to the model's interpretation
Making AI accessible to non-technical users	Recipes abstract away prompt engineering entirely. The creative brief is a structured form; the prompt is generated code the user never sees
API key security in a browser-only app	Thin Bun proxy with JWT validation before every AI call. Keys never exposed to client. SSRF protections on all URL-accepting endpoints
Performant data without loading states	In-memory cache initialised synchronously from localStorage at login. Supabase hydration patches the cache in the background — no visible latency

Relevance to Apple

Pluribus maps directly onto several of Apple's product priorities:

WHAT PLURIBUS DEMONSTRATES

APPLE PRODUCT PARALLELS

- End-to-end product ownership: concept → architecture → UX → deployment
- Human-centred AI design — models as invisible infrastructure
- Privacy-first engineering (data isolation, key security, RLS)
- Performance without compromise — sub-second UI over slow async operations
- Taste-driven design in a dark, high-density visual product

- Apple Intelligence — on-device AI that enhances without interrupting
- Photos app — identity awareness, person recognition, curation at scale
- Final Cut Pro — professional creative tools made accessible
- iCloud — seamless sync, local-first, background persistence
- App Store content tools — structured creative systems for non-engineers

The same design principle that makes Pluribus work — removing technical complexity from creative workflows while preserving full power for those who want it — is the same principle behind Apple's best products. I built Pluribus to prove that principle at scale.